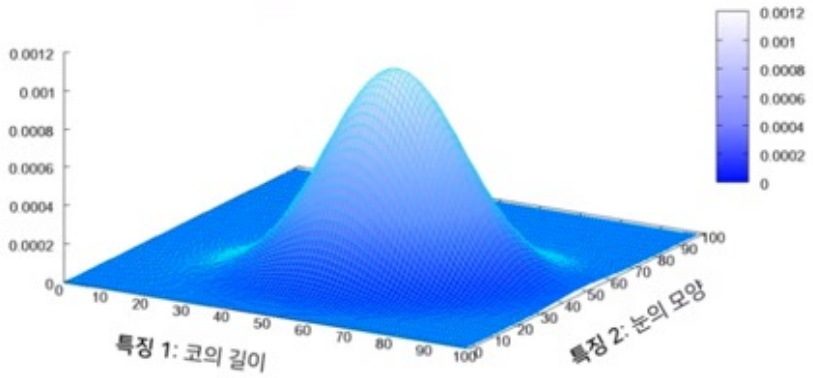
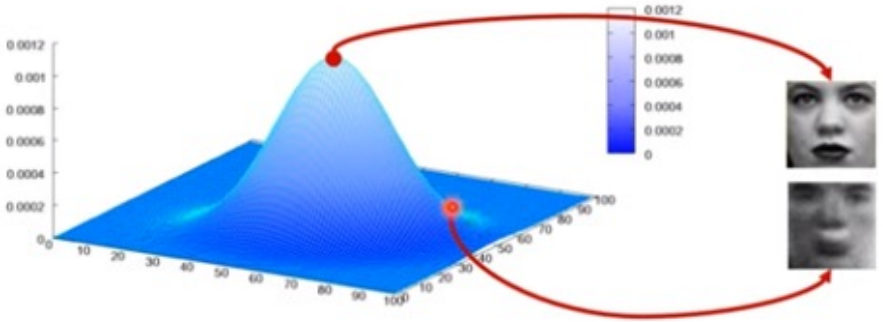
VAE 논문 리뷰

인공지능전공 박현아

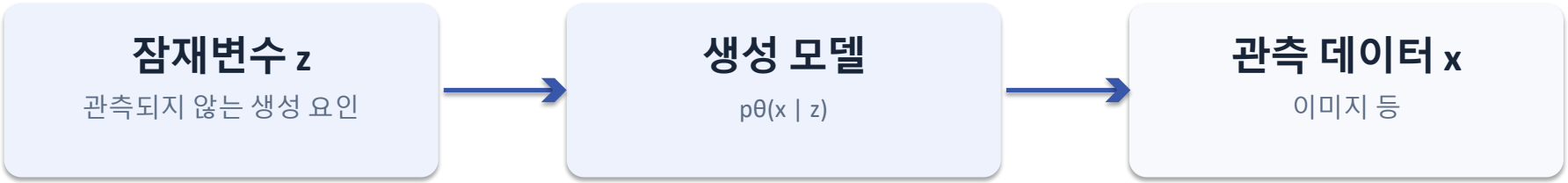


입력 데이터의
분포를 잘 근사하는 모델을 생성



문제: 잠재변수의 사후분포를 계산하기 어렵다

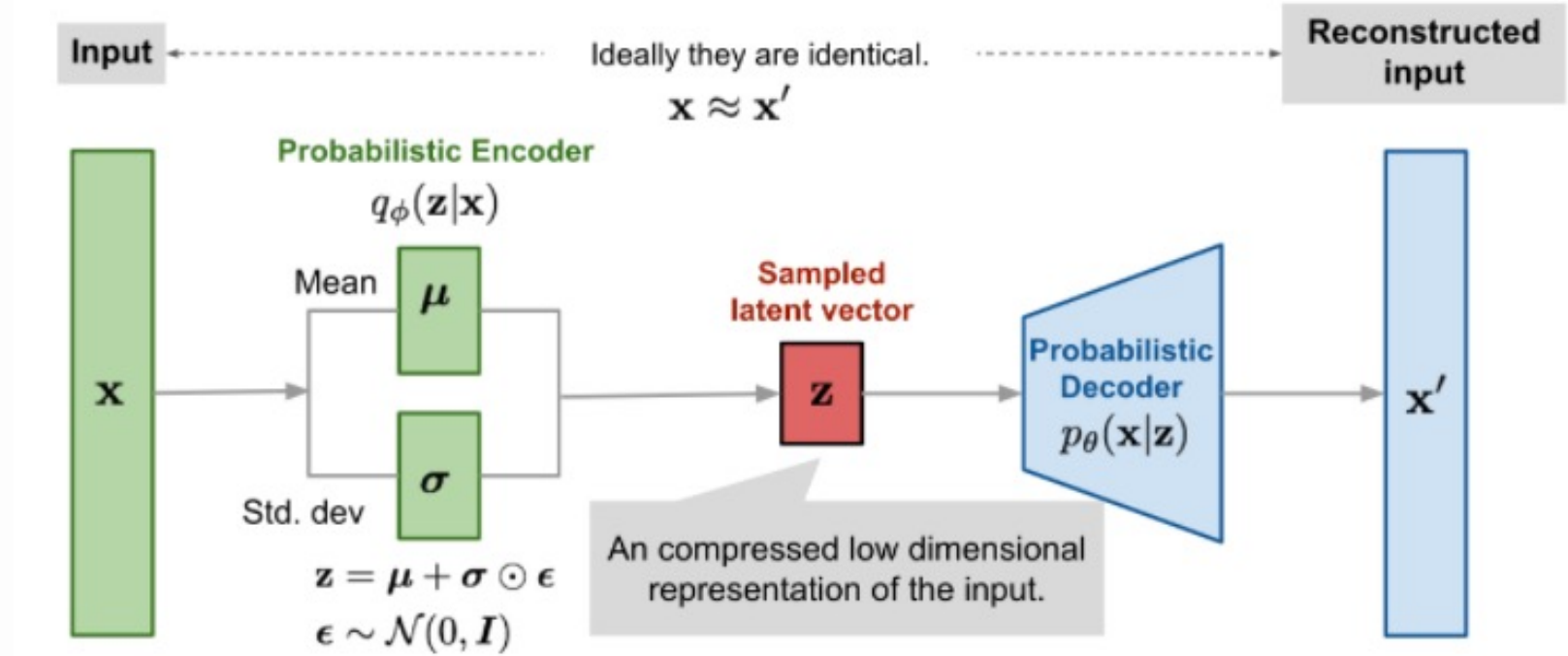
Probabilistic generative models with continuous latent variables



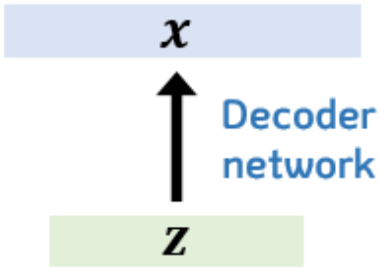
관측한 x에서 "어떤 z가 만들었는가?"를 알려면

$p_\theta(z \mid x)$ → 복잡한 적분 때문에 계산이 어려움

연구 목표: 대규모 데이터에서도 사후 추론과 생성 모델 학습을 효율적으로 수행


Input
Ideally they are identical.
$\mathbf{x} \approx \mathbf{x}'$
Reconstructed input
Probabilistic Encoder
$q_\phi(\mathbf{z}|\mathbf{x})$
Mean
$\boldsymbol{\mu}$
$\boldsymbol{\sigma}$
Std. dev
$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}$
$\boldsymbol{\epsilon} \sim \mathcal{N}(0, \boldsymbol{I})$
$\mathbf{x}$
Sampled latent vector
$\mathbf{z}$
An compressed low dimensional representation of the input.
Probabilistic Decoder
$p_\theta(\mathbf{x}|\mathbf{z})$
$\mathbf{x}'$

Decoder



z latent variable의 확률분포

$p_\theta(z)$

z가 given일 때 x의 확률분포

$p_\theta(x|z^{(i)})$

어떻게 학습?

네트워크의 출력값이 있을 때
우리가 원하는 정답 x가 나올 확률이 높길바람

= **x의 likelihood를 최대화**하는 확률분포 찾자

Maximize

$$p_\theta(x) = \int p_\theta(z)p_\theta(x|z)dz$$

- **Intractability of VAE**

Intractable to compute $p_\theta(x|z)$ for every z

$$p_\theta(\boldsymbol{x}) = \int p_\theta(\boldsymbol{z}) p_\theta(\boldsymbol{x}|\boldsymbol{z}) dz$$

Simple Gaussian prior — Decoder network

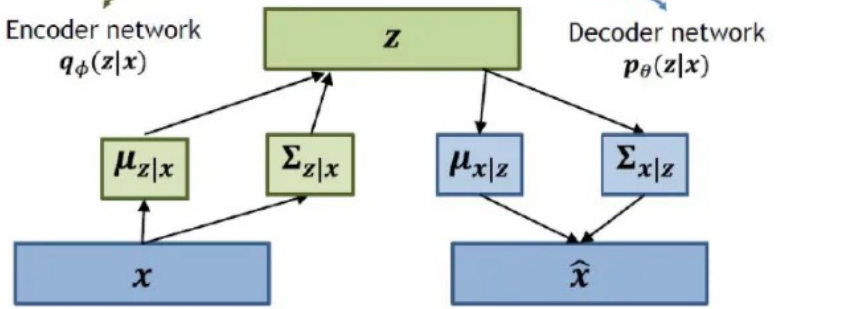
$$p_\theta(\boldsymbol{x}) = \frac{p_\theta(\boldsymbol{x}|\boldsymbol{z}) p_\theta(\boldsymbol{z})}{p_\theta(\boldsymbol{z}|\boldsymbol{x})}$$

Define additional encoder network $q_\phi(\boldsymbol{z}|\boldsymbol{x})$ that approximates $p_\theta(\boldsymbol{z}|\boldsymbol{x})$

This allows us to derive a lower bound on the data likelihood that is tractable, which we can optimize

$$\mathcal{L}(\boldsymbol{x}^{(i)}, \boldsymbol{\theta}, \boldsymbol{\phi}) = \mathbb{E}_{\boldsymbol{z}}\left[\log p_\theta(\boldsymbol{x}^{(i)}|\boldsymbol{z})\right] - D_{KL}(q_\phi(\boldsymbol{z}|\boldsymbol{x}^{(i)})||p_\theta(\boldsymbol{z}))$$

Reconstruct the input data — Make approximate posterior distribution close to prior



핵심 아이디어: Prior와 Encoder는 역할이 다르다

$p(z)$: 생성의 출발점 / $q_\phi(z \mid x)$: 이미지별 근사 사후분포

Prior $p(z)$

사전에 정한 공통 분포

$p(z) = N(0, I)$

새 이미지를 생성할 z를 샘플링

Encoder $q_\phi(z \mid x)$

입력 이미지마다 다른 분포

평균 $\mu(x)$, 표준편차 $\sigma(x)$

계산하기 어려운 $p_\theta(z \mid x)$를 근사

Generative direction $z \rightarrow$ Decoder $\rightarrow x$

Inference direction $x \rightarrow$ Encoder $\rightarrow q_\phi(z \mid x)$

ELBO: 복원 성능과 잠재분포의 균형



Log-likelihood의 계산 가능한 하한을 최대화

ELBO = E qϕ(z|x) [log pθ(x | z)] − KL(qϕ(z | x) || p(z))

① Reconstruction

z로 원본 이미지 x를 잘 설명하기

② KL Divergence

qϕ(z|x)가 공통 p(z)에서 멀어지지 않게

핵심: 모든 이미지의 분포를 똑같게 만드는 것이 아니라, 복원과 규제의 균형을 학습

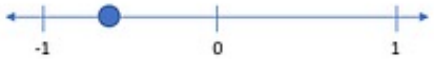


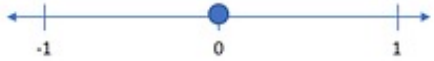


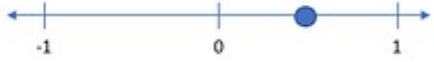


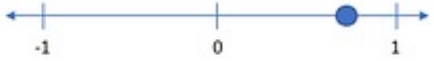


Smile (discrete value)



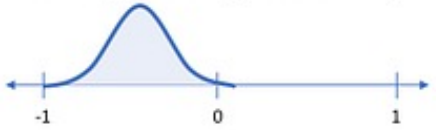


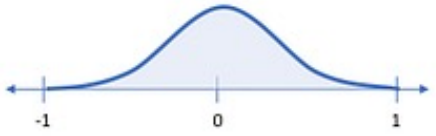


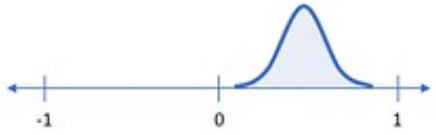


vs.

Smile (probability distribution)









Reparameterization Trick: 무작위성을 분리한다

샘플링이 들어간 모델도 Backpropagation으로 학습

기존: $z \sim q_\phi(z \mid x)$

↓

변경: $z = \mu_\phi(x) + \sigma_\phi(x)\,\varepsilon$

$\varepsilon \sim N(0, I)$

ε는 파라미터 ϕ와 독립적으로 샘플링

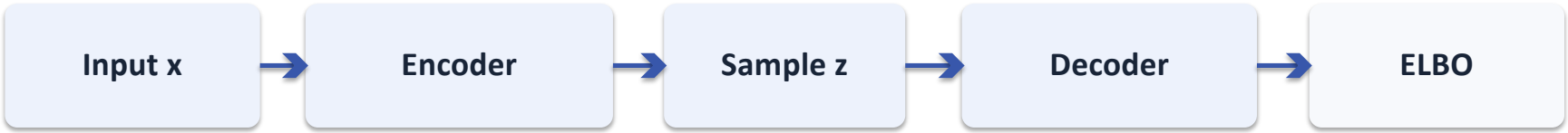
z는 μϕ(x), σϕ(x)의 미분 가능한 함수 → Encoder까지 Gradient 전달

같은 확률분포를 유지하면서 학습 경로를 바꾸는 것이 핵심

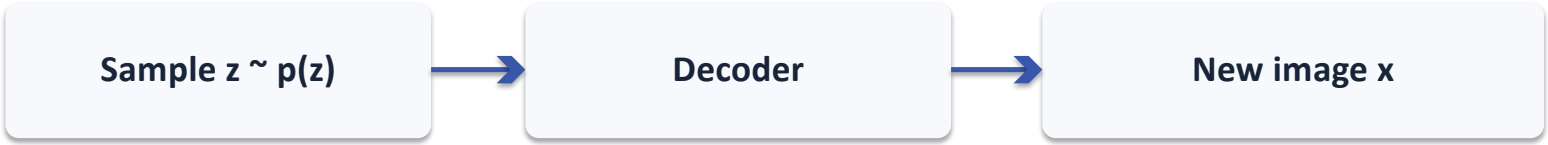
학습은 Encoder+Decoder, 생성은 Decoder만

SGVB estimator → AEVB algorithm → Neural-network VAE

학습 (Training)



새 이미지 생성 (Generation)



실험: MNIST·Frey Face에서 기존 Wake-Sleep 대비 빠른 수렴과 높은 ELBO 확인

참고자료
https://gaussian37.github.io/dl-concept-vae/
https://wikidocs.net/152474